

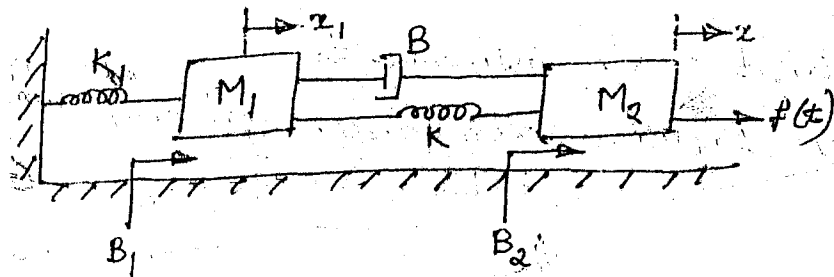
B. Tech Degree VI Semester (Supplementary) Examination September 2010

ME 603 INSTRUMENTATION THEORY AND CONTROL ENGINEERING (2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) Explain zero order systems using a suitable example. (7)
 (b) Derive the equations for time response of a first order system when subjected to (i) unit step input (ii) unit ramp input. Draw response curves. (13)
- OR**
- II. Explain various systematic errors giving suitable examples. Explain the measures taken to minimize these errors. (20)
- III. (a) Explain the method of calibration of strain gauges. (8)
 (b) Explain the working of :
 (i) Rectangular strain gauge rosettes (ii) Delta type strain gauge rosettes. (12)
- OR**
- IV. (a) Explain ORSAT's apparatus with neat sketches. (10)
 (b) Explain with sketches, the principle of a sound level meter. (10)
- V. (a) Define transfer function. (4)
 (b) Write the differential equations governing the mechanical system shown in figure and determine the transfer function. (16)

**OR**

- VI. Find the steady state error when the input is (i) unit step signal (ii) unit ramp signal (iii) unit parabolic signal (20)
- VII. Explain Routh Hurwitz criterion. (20)
- OR**
- VIII. Explain the concept of Liaponov's stability analysis. (20)
- IX. Explain the concept of pneumatic control system with a suitable example. (20)
- OR**
- X. Write short notes on: (i) AC servomotor (ii) Tacho generator. (20)